A model predictive control approach for Qube Servo

Áron Dömötör, Kohlhéb

***Abstract*—This paper studies the effectiveness of Model Predictive Control (MPC) applied for the Qube Servo set up. The base case MPC is established with a novel approach, where the terminal set characterized by the ellipsoid of P is found without the use of the Discrete Algebraic Riccati Equations, but an inherent optimization procedure.**

I. Model

The application is characterized by Figure 1 where two rotatory motions take place orthogonal to each other. The first arm is rotating horizontally, while the second one is aimed to be stabilized at the unstable position of $\alpha = 0$.The states of the system are described as the angles and their angular velocities, while the input is described by the voltage Equation 1. The model is derived based on the principle of least action, resulting in the non-linear model Equation 2

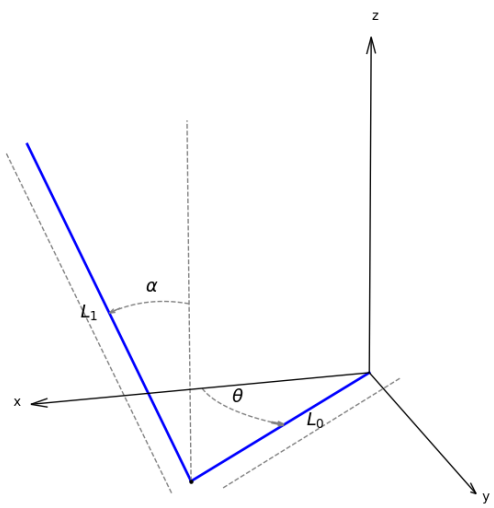

Fig. 1. Model Definition

$$x = \begin{pmatrix}\dot\theta & \dot\alpha & \theta & \alpha\end{pmatrix}^\top \qquad u = V \tag{1}$$

$$\begin{pmatrix}\ddot\theta\\ \ddot\alpha\end{pmatrix} = \bar M^{-1}\left(\frac{1}{L_1^2 m_1}\begin{pmatrix}\tau_\theta\\ \tau_\alpha\end{pmatrix} + \frac{g}{2L_1}\begin{pmatrix}0\\ \sin(\alpha)\end{pmatrix}\right) + \bar M^{-1}\frac{1}{2}\begin{pmatrix}-2\dot\alpha & 0\\ \dot\theta & -\dot\alpha\end{pmatrix}\frac{\partial M}{\partial\alpha}\begin{pmatrix}\dot\theta\\ \dot\alpha\end{pmatrix} \tag{2}$$

where $\bar M$ being defined as $M + \begin{pmatrix}1\\0\end{pmatrix} J_0/(L_1^2 m_1)\begin{pmatrix}1 & 0\end{pmatrix}$, with M being defined as

$$M := \tag{3}$$

$$\begin{pmatrix} r^2 C_0^\top C_0 + rC_0^\top C_1 + \frac{1}{3}C_1^\top C_1 & \frac{1}{2}rC_0^\top C_2 + \frac{1}{3}C_1^2 \\ \frac{1}{2}rC_0^\top C_2 + \frac{1}{3}C_1^2 & \frac{1}{3}C_2^\top C_2 \end{pmatrix} \tag{4}$$

Áron Dömötör, Kohlhéb is a Systems and Control master student at TU Delft, The Netherlands. E-mail address: {Kohlheb-1}@student.tudelft.nl.

Where C_0, C_1, C_2 are defined as

$$\begin{pmatrix}C_0\\C_1\\C_2\end{pmatrix} = \left(\begin{array}{c} -sin(\theta)\\ cos(\theta)\\ 0 \\ \hline sin(\alpha)cos(\theta)\\ sin(\alpha)sin(\theta)\\ 0 \\ \hline cos(\alpha)sin\theta)\\ -cos(\alpha)cos(\theta)\\ -sin(\alpha)\end{array}\right) \tag{5}$$

And the torque is defined by the sum of the gravity, the cable acting as spring, engine, and friction:

$$\frac{1}{L_1^2 m_1}\begin{pmatrix}\tau_\theta\\ \tau_\alpha\end{pmatrix} = \begin{pmatrix}-k1 & 0\\ 0 & -k_2\end{pmatrix}\begin{pmatrix}\dot\theta\\ \dot\alpha\end{pmatrix} + \tag{6}$$

$$\begin{pmatrix}-k_3 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}\theta\\ \alpha\end{pmatrix} + \begin{pmatrix}k_4\\ 0\end{pmatrix}V \tag{7}$$

This leads to the linearization that is identified via measurements with the shape

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14}\\ a_{21} & a_{22} & a_{23} & a_{24}\\ 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\end{pmatrix} \qquad B = \begin{pmatrix}b_1\\ b_2\\ 0\\ 0\end{pmatrix} \tag{8}$$

$$C = \begin{pmatrix}0 & I_{2\times 2}\end{pmatrix} \qquad D = 0 \tag{9}$$

The system is sampled at sampling time h, therefore the model is discretized as Equation 10

$$x(k+1) = e^{Ah}x(k) + (e^{Ah} - I)A^{-1}Bu(k) =: Fx(k) + Gu(k) \tag{10}$$

Furthermore the non-linear model is also discretized using the forward Euler method

$$x(k+1) = f^+(x(k), u(k)) := \int_0^h \dot x\, dt \approx h\,\dot x \tag{11}$$

II. MPC Definition

With finite horizon N the MPC problem is formulated as the optimization problem to find the optimal input sequence $u_{0:N-1}$

$$\mathrm{P_N}: \qquad \min_{\bar u_{N-1}\in\mathbb{U}} V_N(x_0, u_{0:N-1}) \tag{12}$$

$$s.t. \qquad \phi_N(x_0, \bar u_{N-1}) \in \mathbb{X}_f \tag{13}$$

$$g(u_{0:N-1}) \le 0 \tag{14}$$

$$0 = u_{0:N-1} - K_{0:N-1}x_{0:N-1} \tag{15}$$

$$x(k) \in \mathbb{X}\ \forall\, k \in \mathbb{Z} \cup [0, N] \tag{16}$$

where ϕ_k can be expressed as Equation 17 for the linear state feedback case, and Equation 18 for the general case.

$$\phi_k(x_0, \bar K_{k-1}) = \left[\prod_{i=0}^{k-1} F - GK_i\right]x_0 \tag{17}$$

$$\phi_k(x_0, \bar u_{k-1}) = F^k x_0 + \sum_{i=0}^{k-1} F^{K-1-i}Gu_i \tag{18}$$

The loss function consists of a quadratic stage, or running cost and a quadratic terminal cost Equation 19

$$V_N = x_{0:N-1}^\top Q_{N-1}^e x_{0:N-1} + u_{0:N-1}^\top R_{N-1}^e u_{0:N-1} + x_N^\top P_f x_N \tag{19}$$

where the $x_{0:N-1}, Q_{0:N-1}, R_{0:N-1}$ are defined as

$$x_{0:N-1} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ .. \\ x_{N-1} \end{pmatrix} = \begin{pmatrix} I \\ F - GK_0 \\ (F-GK_1)(F-GK_0) \\ .. \\ \prod_{i=0}^{N-2}(F-GK_i) \end{pmatrix} x_0 \tag{20}$$

$$Q_{0:N-1} = I_{N-1} \otimes Q \tag{21}$$

$$u_{0:N-1} = \begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ .. \\ u_{N-1} \end{pmatrix} = -K_{0:N-1} x_{0:N-1} \tag{22}$$

$$K_{0:N-1} = \operatorname{diag}\begin{pmatrix} K_0 & K_1 & .. & K_{N-1} \end{pmatrix} \tag{23}$$

$$u_{0:N-1} = -\begin{pmatrix} K_0 \\ K_1(F-GK_0) \\ K_2(F-GK_1)(F-GK_0) \\ .. \\ K_{N-1}\prod_{i=0}^{N-2}(F-GK_i) \end{pmatrix} x_0 \tag{24}$$

Furthermore $\mathbb{X}$ is the state space and is defined as Equation 25

$$\mathbb{X} = \mathbb{X}_{\dot\theta} \times \mathbb{X}_{\dot\alpha} \times \mathbb{X}_\theta \times \mathbb{X}_\alpha \tag{25}$$

$$with \quad \mathbb{X}_{\dot\theta=[-\dot\theta_{max},\dot\theta_{max}]} \tag{26}$$

$$\mathbb{X}_{\dot\alpha=[-\dot\alpha_{max},\dot\alpha_{max}]} \tag{27}$$

$$\mathbb{X}_\theta = [-\theta_{max}, \theta_{max}] \tag{28}$$

$$\mathbb{X}_\alpha = [-\alpha_{max}, \alpha_{max}] \tag{29}$$

and $\mathbb{X}_f$ is the terminal set defined in the stability discussion section III

III. Stability

MPC incorporates a receding horizon approach, where the finite horizon trajectory is computed at each time step and the first element of the optimal control sequence is applied. The stability of the process can be proven by the stability of the finite horizon trajectory, since the receding horizon solution compounds of the finite horizon results.

Stability is proven by using Lyapunov functions to bound the evolution of the state. Given the Lyapunov function meets the properties: it is strictly increasing in x, it is strictly decreasing in $t = kh$ they prove local stability. Since quadratic expressions are always positive if their weight matrix is by definition positive definite the property is utilized for $Q, R, P_f \succ 0$. Therefore the cost function of the form Equation 19 meets the first requirement.

The second required property of the cost function over trajectory is divided into two parts: the stage part and the post horizon part. Given during the stage part the control input $K(k)$ is applied.

$$K(k) = (B^\top \Pi(k+1)G + R)^{-1} G^\top \Pi(k+1) F \tag{30}$$

where $\Pi(k)$ is given as

$$\Pi(k) = Q + F^\top \Pi(k+1)(F + GK(k)) \tag{31}$$

This solution is proven to result in a strictly decreasing V in t for an infinite horizon case, therefore it proves stability given that the tail of the sequence is also stable.

The stability of the post horizon phase is given for applying a constant control gain $K(N-1)$ and is characterized by the solution to it. For the last step of the horizon the solution to DARE gives the result Equation 32.

$$K(N-1) = (G^\top P_f G + R)^{-1} G^\top P_f F \tag{32}$$

which results in the dynamics

$$x(k+1) = \left(F - G(G^\top P_f G + R)^{-1} G^\top P_f F\right) x(k) \tag{33}$$

The evolution of the dynamics is bounded only if $x(k)$ is part of the terminal set defined as Equation 34. Here the right side needs to be some constant, it doesn't hold significance as it only scales the entries of P_f, therefore it can be chosen to be unit it doesn't restrict the solution space.

$$\mathbb{X}_f := \{x\} \; \forall \; x | \; x^\top P_f x \leq 1 \tag{34}$$

Since DARE guarantees the reduction of $x^\top P_f x$ in t, $\mathbb{X}_f$ is positive invariant with respect to the dynamics Equation 33, meaning that all possible evolutions of x are contained in $\mathbb{X}_f$.

However, $\mathbb{X}_f$ needs to be reached within N from the initial state x_0, which is not always possible since u is constrained. Thus it creates a requirement on x_0: x_0 needs to be in $\mathbb{X}_N$ with Equation 35

$$\mathbb{X}_N = \{x\} \; \forall \; x \; | \exists \; u_{0:N-1} \; s.t. \phi_N(x, u_{0:N-1}) \; \in \; \mathbb{X}_f \tag{35}$$

Assuming the system is linear the optimal value function can be used as the Lyapunov function, meaning that for stability:

$$V_f(x(k+1)) - V_f(x(k)) \leq 0 \tag{36}$$

which can be expanded with the dynamics to an expression of a matrix that needs to be semi negative definite.

$$(F - GK(N-1))^\top P_f (F - GK(N-1)) - P_f \preceq 0 \tag{37}$$

In order to guarantee asymptotic stability, the reduction in the value function should be lower bounded by a strictly positive function. Since the running cost consists of quadratics with positive semi-definite matrices, with either Q or R having to be non-zero, the running cost can serve as the bounding function for the decrease:

$$V_f(x(k+1)) - V_f(x(k)) \leq -\ell(x(k), u(k)) \tag{38}$$

which results in the following matrix inequality that must be negative semi-definite:

$$(F - GK(N-1))^\top P_f (F - GK(N-1)) - P_f \tag{39}$$

$$+Q + K(N-1)^\top R K(N-1) \preceq 0 \tag{40}$$

A. Non-linear Stability

Since the model used is not an LTI system, it is only treated as one for the controller design, one additional condition has to be satisfied. Equation 41 needs to hold for $x \in \mathbb{X}_f$. Since $\mathbb{X}_f \subset \mathbb{X}$ if this condition is met for all $x \in \mathbb{X}$, it is also met for all $x \in \mathbb{X}_f$. For this reason, when the MPC is constructed the corner points are of $\mathbb{X}$ are checked. Since $\mathbb{X}$ is a convex polytopic set by construction, the rest of the points can be constructed as the convex combination of the corner points. This step is done by a numeric check.

$$V_f(x(k+1)) - V_f(x(k)) \leq 0 \tag{41}$$

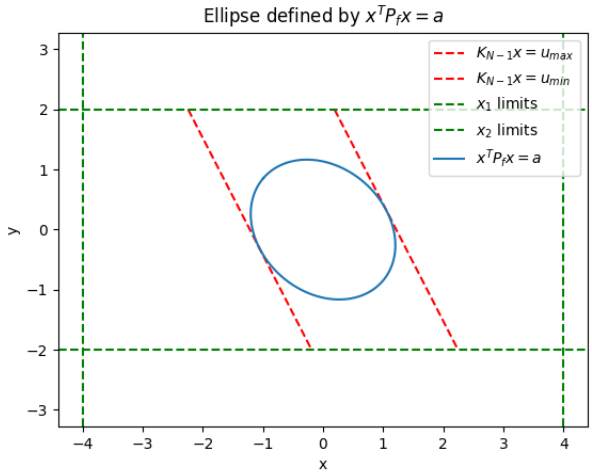

Fig. 2. P_f 2D example

IV. Finding P_f

For the terminal set to comply with the optimization problem Equation 12 $\mathbb{X}_f$ needs to meet two additional requirements

$$\mathbb{X}_f \subset \mathbb{X} \tag{42}$$

$$-K(N-1)x \in \mathbb{U} \, \forall \, x \in \mathbb{X}_f \tag{43}$$

Which is reformulated as

$$(-K(N-1)x)_{max} \in \mathbb{U} \mid x \in \mathbb{X}_f \tag{44}$$

$$(x_i)_{max} \in \mathbb{X}_f \tag{45}$$

Since $\mathbb{X}_f$ is a convex set, finding the maximum is analogous to finding the boundary that is orthogonal to the gradient of the optimized function, therefore this can be reformulated as Equation 46 where e_i is the i-th directional unit vector and where $P_f^{-1}(ii)$ stands for the i-th diagonal entry of the matrix P_f^{-1}.

$$-K(N-1)x| \; K(N-1) \perp \partial \mathbb{X}_f = \tag{46}$$

$$K_{N-1}\sqrt{a}\frac{aP_f^{-1}K_{N-1}^\top}{\sqrt{K_{N-1}P_f^{-1}K_{N-1}^\top}} \tag{47}$$

$$x_i \mid e_i \perp \partial \mathbb{X}_f = \sqrt{aP_f^{-1}(i,i)} \tag{48}$$

In order to increase the performance of the MPC $\mathbb{X}_f$ needs to be as large as possible, this is formulated as optimizing for the maximum volume of the ellipsoid that defines $\mathbb{X}_f$, which is inversely proportional to the determinant of P_f, therefore the determinant of P_f is minimized for in the process.

There are multiple ways to find P_f such that all constraints are met, but computationally it gets complicated. Finding it through DARE offers a simple computation, where the shape of P_f is first found by Equation 49, and then the matrix is scaled such that the terminal set meets the conditions Equation 42.

$$P_f = F^\top P_f F - F^\top P_f G(R + G^\top P_f G)^{-1} G^\top P_f F + Q \tag{49}$$

The result of this process is visualized for the $\mathbb{X} \in \mathbb{R}^2$ case

V. Base Case - Numerical Tests

In this section the evolutions are plotted with the initial state of $x_0 = \big(0, 0, -\beta, \beta\big)$, $\beta \geq 0$. This has been chosen due to the dynamics of the system. To maintain stability of the system for a positive α $\dot{\theta}$ needs to be negative. This poses the underlying challenge to the controller, if the initial angles are both positive the controller has a very easy task to drive both of them to 0.

This process with a sampling time of $h = 0.01$, horizon of $N = 40$, $Q = diag(0, 0, 0, 1)$ and $R = 1000$ it produces evolution Figure 3. Under high horizon, as expected, the controller acts very similar to an LQR controller Figure 5 . However, for a low R value this is not the case, here the limitations of u make the convergence of MPC slower and the model predictive element makes MPC oscillate less as well. For a very low horizon of $N = 3$, but still high $R = 1000$ MPC differs from LQR in a different way Figure 6. In this case MPC is performing worse, which corresponds to the trade of of high N better controller but worse performance.

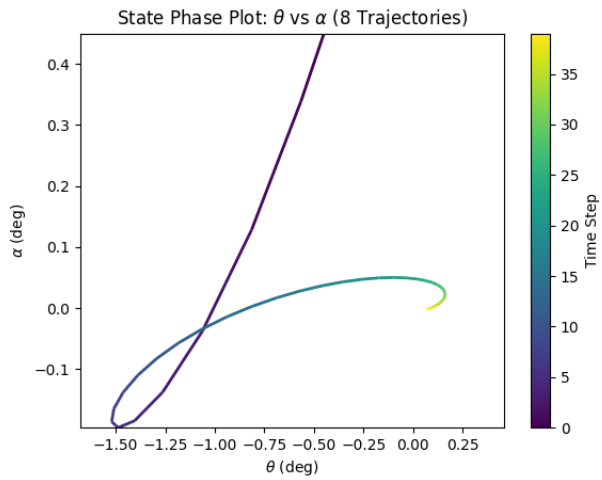

Fig. 3. Evolution: high R, high N

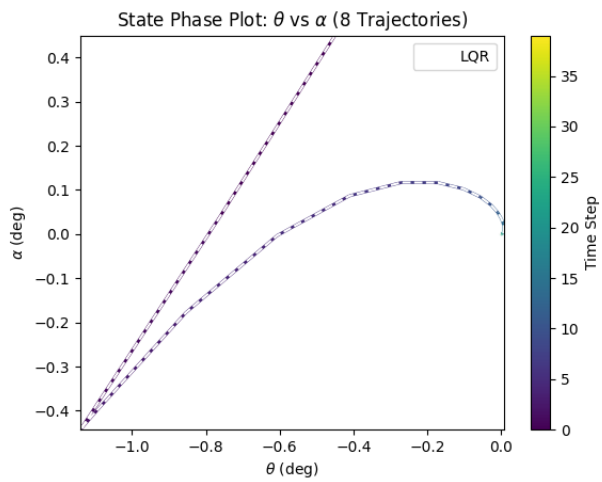

Fig. 4. LQR & MPC comparison: high R, high N

VI. Assumptions and Design Choices

- The system is sampled with a constant sampling time $T_s = h$. [Met by definition]
- Zero-order hold (ZOH) is applied to the control input between samples.[Met by definition]
- The control input is linear with respect to the state: $u_k = -Kx_k$.[Met by definition]
- The system is sufficiently modelled as a Linear Time-Invariant (LTI) system.[Met by testing]
- The system is both observable and controllable. [Met by definition]

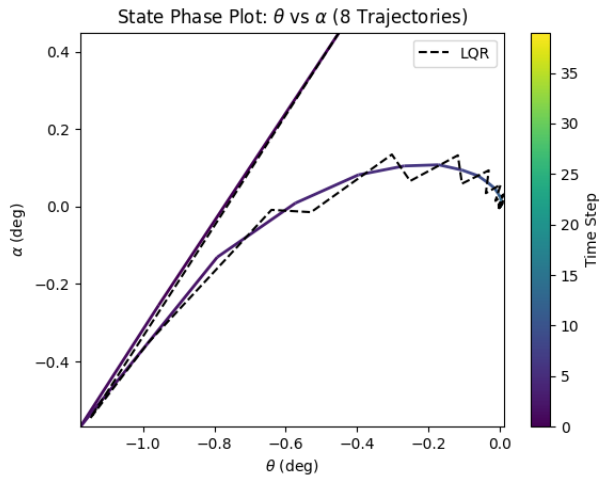

Fig. 5. LQR & MPC comparison: low R, high N

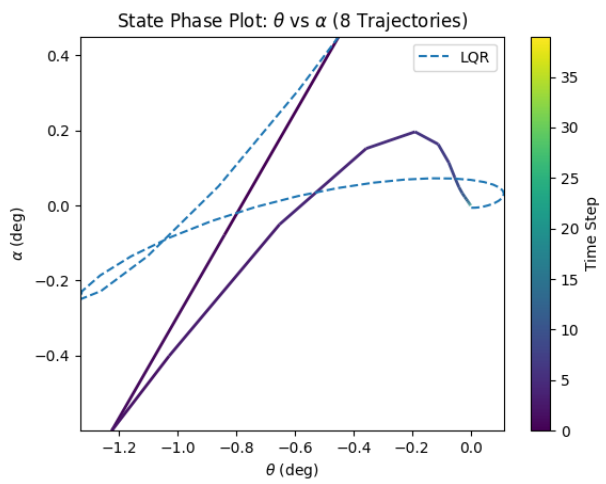

Fig. 6. LQR & MPC comparison: high R, low N

- Measurements are noise-free and contain no offset.
- The plant dynamics satisfy $f(0,0) = 0$. [Met by defining the system such that the stable point is the origin]
- The admissible state set $\mathbb{X}$ and input set $\mathbb{U}$:[Met by definition]
  - are closed, bounded (hence compact), and time-invariant,
  - include the origin, i.e., $0 \in \mathbb{X},\ 0 \in \mathbb{U}$,
  - are defined by boundaries corresponding to the 1-norm.
  - are convex
- The stage cost is quadratic: $\ell(x,u) = u_k^T R u_k + u_k^T R u_k$, with $R \succ 0$.[Met by definition]
- The terminal cost is quadratic: $V_f(x) = x_N^T P x_N$, with $P \succ 0$.[Met by definition]
- The cost functions satisfy:[Met by definition]
  - $V_f(0) = 0$, $\ell(0,0) = 0$, and both are continuous,
  - There exist class-$\mathcal{K}$ functions c_1, c_2 such that:

$$c_1(\|x\|) \le V_f(x) \le c_2(\|x\|), \quad c_1(\|(x,u)\|) \le \ell(x,u) \le c_2(\|(x,u)\|).$$

- A stabilizing control law $\kappa(x) = Kx$ is defined over the terminal region $\mathbb{X}_f \subset \mathbb{R}^n$.[Met by definition of K]
- The closed-loop matrix $F + GK$ is Schur stable.[Met by definition of K]
- The terminal cost satisfies the discrete-time Lyapunov inequality:[Met by constructing P_f and K_{N-1} with DARE]

$$(F+GK)^T P_f (F+GK) - P_f \prec -Q - K^T R K.$$

- The decrease condition holds:[Met by testing]

$$V_f(f(x,Kx)) \le V_f(x) - \ell(x,Kx), \quad \forall x \in \mathbb{X}_f.$$

VII. Model Parameters

For reproducibility purposes the model has the parameters:

TABLE I
Numerical values for system matrices and parameters

Parameter	Value
r	1.00002879
k_0	0.57272953
k_1	0.09286886
k_2	0.19999285
k_3	9.99793729
k_4	949.857911
k_5	76.0387603

$$A = \begin{pmatrix} -0.11288 & -0.36463 & -12.15194 & 69.31770 \\ -0.16932 & -1.14694 & -18.22844 & 218.03768 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$B = \begin{pmatrix} 1154.50006 \\ 1731.79996 \\ 0 \\ 0 \end{pmatrix}$$